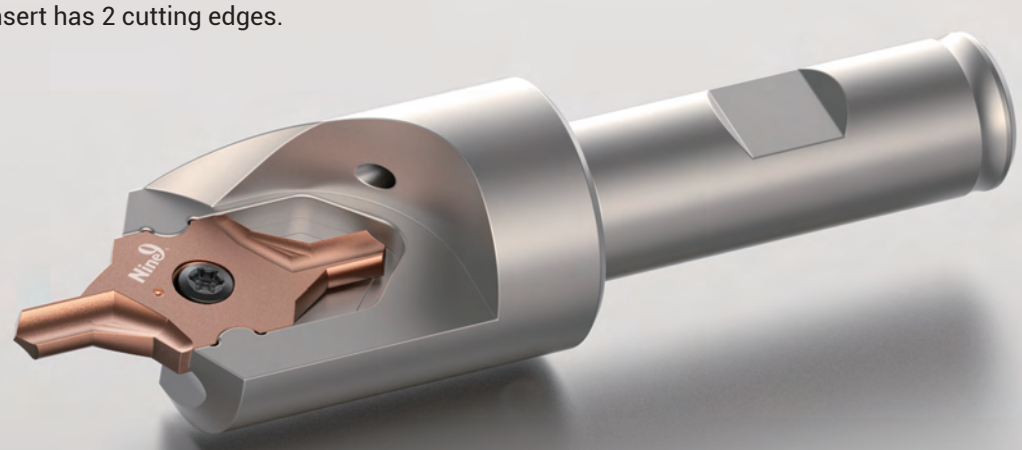
# i-Center®

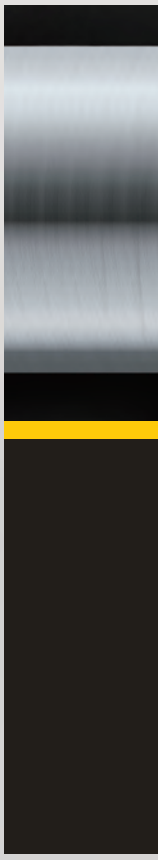
# Center Drill >>>

The " i-Center " is a trademark of Nine9,
the developer of the first indexable center drill in the world.(Patented)
Offering an indexable insert system for the 1st time, Nine9's "i-Center " design improves your process performance.

P M K N

- 2 cutting flutes design same as carbide center drill for high performance speed and feed rate.
- Each insert has 2 cutting edges.





## Features >>>

### High Speed, High Feed Rate

- The special ground insert and rigid holder design facilitate high performance speed and feed rates. For example, drilling alloy steel at 6000 rpm and feed rate of 600 mm/min. (0.1 mm/rev.)

### Excellent Repeatability

- The positioning repeatability of the insert is within 0.02 mm (.0008")
  in radial direction, thus ensuring conformity to any national standards.

### Easy Tool Length Setting

- The axial position accuracy of the insert is 0.05 mm (.002"). It is not necessary to reset the tool length when changing the insert or cutting edge.

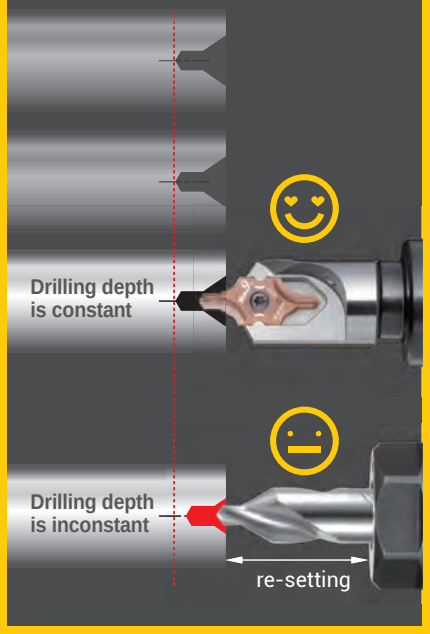
### Extended Tool Life

- Coolant can be supplied through the center of the holder to increase performance and extend tool life.
- Insert geometry, grades and coating process are specifically engineered for centering applications.

# Applications



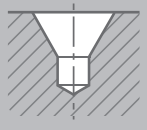
Excellent repeatability
No need tool length re-setting while changing insert or cutting edge.



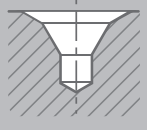

**DIN 332 Form R**
Ø1.0~Ø10



**DIN 332 Form A**
Ø2.0~Ø3.15



**DIN 332 Form A+B**
Ø1.0~Ø10



**ANSI 60°**
#2.0~#10



"World's first indexable center drill. Shortens set up and center drilling time. Increases tool life and reduces tooling costs."



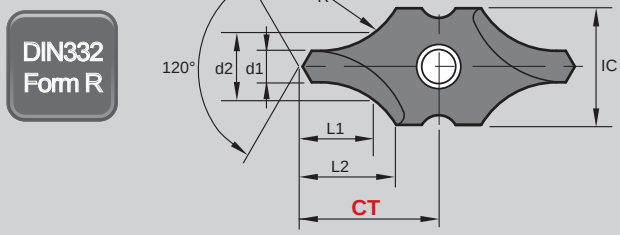
• Internal coolant can be supplied to the tip of center drill insert.

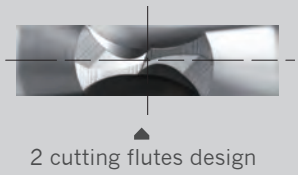
**NC2057** 
- P35 grade, AlTiN + TiSiN coating, Universal grade for all kind of steel.
- Double-edged cutting, fully ground insert for improving machining stability. ( IC10 inserts )

**NC5074** 
- P40 grade, Helica (AlCrN) coating, design for small diameter center drill ( IC08 inserts ).

**NC2033** 
- K20F grade, TiAlN coated, for carbon steel, alloy steel, high alloy steel and cast iron.

# DIN332 Form R




2

i-Center



**For DIN332 Form R Center Hole >>**

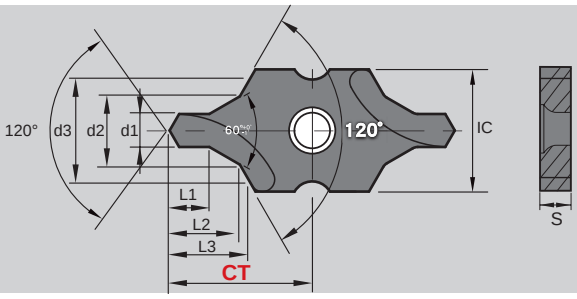
| IC | Code | Parts No. | Coating | Grade | d1 | | d2 | L1 | L2 | R | S | CT ±0.025 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 08 | 032211 | I9MT08T1R0100-NC5074 | Helica | P40 | 1.00 | + 0.14 0 | 2.12 | 2.16 | 4.14 | 2.8 | 2.00 | 7.55 |
| | 032212 | I9MT08T1R0125-NC5074 | | | 1.25 | | 2.65 | 2.74 | 4.64 | 3.5 | | 7.90 |
| | 032213 | I9MT08T1R0160-NC5074 | | | 1.60 | | 3.35 | 3.45 | 5.13 | 4.5 | | 8.40 |
| | 032214 | I9MT08T1R0200-NC5074 | | | 2.00 | | 4.25 | 4.45 | 6.08 | 5.65 | | 9.10 |
| 10 | 031200 | I9MT1003R0100-NC2057 | AlTiN+ TiSiN | P35 | 1.00 | + 0.14 0 | 2.12 | 2.16 | 4.72 | 2.8 | 3.00 | 12.35 |
| | 031201 | I9MT1003R0125-NC2057 | | | 1.25 | | 2.65 | 2.74 | 5.22 | 3.5 | | |
| | 031202 | I9MT1003R0150-NC2057 | | | 1.50 | | 3.60 | 3.67 | 6.14 | 5.0 | | |
| | 031203 | I9MT1003R0160-NC2057 | | | 1.60 | | 3.35 | 3.45 | 5.32 | 4.5 | | |
| | 031204 | I9MT1003R0200-NC2057 | | | 2.00 | | 4.25 | 4.45 | 6.50 | 5.65 | | |
| | 031205 | I9MT1003R0250-NC2057 | | | 2.50 | | 5.30 | 5.59 | 7.66 | 7.15 | | |
| | 031206 | I9MT1003R0300-NC2057 | | | 3.00 | + 0.18 0 | 5.70 | 6.92 | 9.50 | 10.00 | | |
| | 031207 | I9MT1003R0315-NC2057 | | | 3.15 | | 6.70 | 7.21 | 8.93 | 9.00 | | |
| 12 | 033201 | I9MT12T2R0200-NC2033 | TiAlN | K20F | 2.00 | + 0.14 0 | 4.25 | 4.45 | 6.64 | 5.65 | 2.54 | 11.73 |
| | 033202 | I9MT12T2R0250-NC2033 | | | 2.50 | | 5.3 | 5.59 | 8.11 | 7.15 | | 13.00 |
| | 033203 | I9MT12T2R0315-NC2033 | | | 3.15 | + 0.18 0 | 6.7 | 7.21 | 9.63 | 9.0 | | 14.00 |
| 16 | 034201 | I9MT1603R0400-NC2033 | | | 4.00 | | 8.5 | 9.06 | 12.23 | 11.0 | 3.18 | 19.40 |
| | 034202 | I9MT1603R0500-NC2033 | | | 5.00 | | 10.6 | 11.45 | 14.2 | 14.0 | | 19.40 |
| 20 | 035201 | I9MT2004R0630-NC2033 | | | 6.30 | + 0.22 0 | 13.2 | 14.63 | 18.2 | 18.0 | 4.76 | 28.40 |
| | 035202 | I9MT2004R0800-NC2033 | | | 8.00 | | 17.0 | 18.63 | 20.44 | 22.5 | | 28.30 |
| 25 | 036201 | I9MT2506R1000-NC2033 | | | 10.00 | | 21.2 | 23.51 | 25.8 | 28.0 | 6.35 | 34.20 |

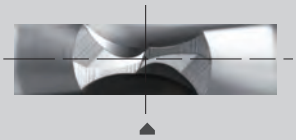
The quantity of insert per box.:

| IC 08 | IC 10 | IC 12 | IC 16 | IC 20 | IC 25 |
|---|---|---|---|---|---|
| 5 | 5 | 5 | 2 | 1 | 1 |

# DIN332 Form A+B

DIN332
Form A+B



2 cutting flutes design



**For DIN332 Form A+B Center Hole >>**



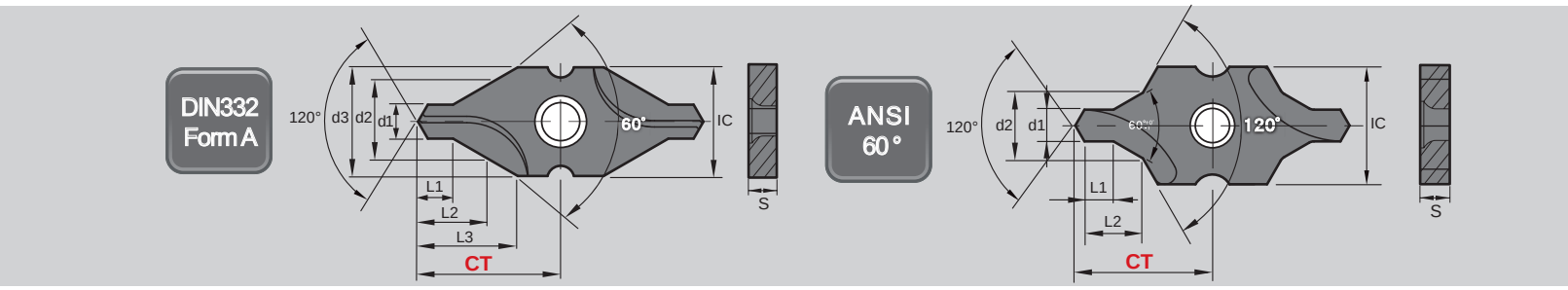
| IC | Code | Parts No. | Coating | Grade | d1 | | d2 | d3 | L1 | L2 | L3 | S | CT ±0.025 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 08 | 032011 | I9MT08T1B0100-NC5074 | Helica | P40 | 1.00 | + 0.14 0 | 2.12 | 3.15 | 1.3 | 2.21 | 2.51 | 2.00 | 7.55 |
| | 032012 | I9MT08T1B0125-NC5074 | | | 1.25 | | 2.65 | 4.0 | 1.6 | 2.75 | 3.14 | | 7.90 |
| | 032013 | I9MT08T1B0160-NC5074 | | | 1.60 | | 3.35 | 5.0 | 2.0 | 3.46 | 3.93 | | 8.40 |
| | 032014 | I9MT08T1B0200-NC5074 | | | 2.00 | | 4.25 | 6.3 | 2.5 | 4.39 | 4.98 | | 9.10 |
| 10 | 031000 | I9MT1003B0100-NC2057 | AlTiN+ TiSiN | P35 | 1.00 | + 0.14 0 | 2.12 | 3.15 | 1.3 | 2.21 | 2.51 | 3.00 | 12.35 |
| | 031001 | I9MT1003B0125-NC2057 | | | 1.25 | | 2.65 | 4.0 | 1.6 | 2.75 | 3.14 | | |
| | 031002 | I9MT1003B0150-NC2057 | | | 1.50 | | 3.18 | 4.50 | 2.0 | 3.45 | 3.84 | | |
| | 031003 | I9MT1003B0160-NC2057 | | | 1.60 | | 3.35 | 5.0 | 2.0 | 3.46 | 3.93 | | |
| | 031004 | I9MT1003B0200-NC2057 | | | 2.00 | | 4.25 | 6.3 | 2.5 | 4.39 | 4.98 | | |
| | 031005 | I9MT1003B0250-NC2057 | | | 2.50 | | 5.3 | 8.0 | 3.1 | 5.53 | 6.28 | | |
| | 031006 | I9MT1003B0300-NC2057 | | | 3.00 | + 0.18 0 | 6.46 | 9.00 | 4.1 | 7.10 | 7.83 | | |
| | 031007 | I9MT1003B0315-NC2057 | | | 3.15 | | 6.7 | 10.0 | 3.9 | 6.90 | 7.85 | | |
| 12 | 033001 | I9MT12T2B0200-NC2033 | TiAlN | K20F | 2.00 | + 0.14 0 | 4.25 | 6.3 | 2.5 | 4.39 | 4.98 | 2.54 | 11.73 |
| | 033002 | I9MT12T2B0250-NC2033 | | | 2.50 | | 5.3 | 8.0 | 3.1 | 5.53 | 6.28 | | 13.0 |
| | 033003 | I9MT12T2B0315-NC2033 | | | 3.15 | + 0.18 0 | 6.7 | 10.0 | 3.9 | 6.90 | 7.85 | | 14.0 |
| 16 | 034001 | I9MT1603B0400-NC2033 | | | 4.00 | | 8.5 | 12.5 | 5.0 | 8.9 | 10.03 | 3.18 | 19.4 |
| | 034002 | I9MT1603B0500-NC2033 | | | 5.00 | | 10.6 | 16.0 | 6.3 | 11.15 | 12.68 | | 19.4 |
| 20 | 035001 | I9MT2004B0630-NC2033 | | | 6.30 | + 0.22 0 | 13.2 | 18.0 | 8.0 | 13.98 | 15.33 | 4.76 | 28.4 |
| | 035002 | I9MT2004B0800-NC2033 | | | 8.00 | | 17.0 | *20 | 10.1 | 17.89 | 18.73 | | 28.3 |
| 25 | 036001 | I9MT2506B1000-NC2033 | | | 10.00 | | 21.2 | *25 | 12.8 | 22.5 | 23.57 | 6.35 | 34.2 |

* Notice: The d3 size is different from DIN332 center hole.

The quantity of insert per box.:

| IC 08 | IC 10 | IC 12 | IC 16 | IC 20 | IC 25 |
|---|---|---|---|---|---|
| 5 | 5 | 5 | 2 | 1 | 1 |

# DIN332 Form A & ANSI 60°








## For DIN332 Form A Center Hole >>

The quantity of insert per box.:

| IC 08 | IC 10 | IC 12 | IC 16 | IC 20 | IC 25 |
|---|---|---|---|---|---|
| 5 | 5 | 5 | 2 | 1 | 1 |

| IC | Code | Parts No. | Coating | Grade | d1 | | d2 | d3 | L1 | L2 | L3 | S | CT ±0.025 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 08 | 032114 | I9MT08T1A0200-NC5074 | Helica | P40 | 2.0 | + 0.03 0 | 4.25 | 8 | 2.15 | 4.10 | 7.35 | 2.00 | 10.5 |
| | 032115 | I9MT08T1A0250-NC5074 | | | 2.5 | | 5.3 | | 2.58 | 5.00 | 7.34 | | |
| | 032116 | I9MT08T1A0315-NC5074 | | | 3.15 | + 0.03 0 | 6.7 | | 3.23 | 6.30 | 7.43 | | |

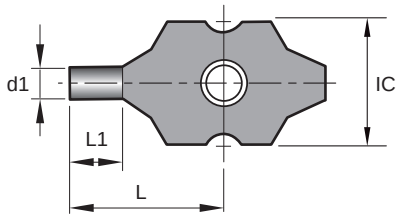


## For ANSI 60° Center Hole >>

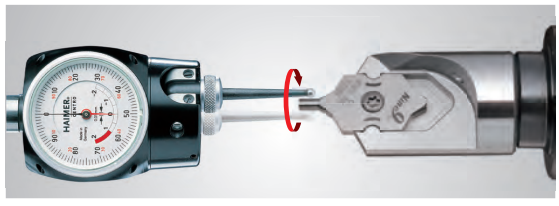
| IC | Code | Parts No. | Coating | Grade | Size | d1 | | | d2 | | L1 | | L2 | S | CT ±0.025 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | mm | | | mm | | mm | mm | | |
| 12 | 033101 | I9MT12T2A2-NC2033 | TiAlN | K20F | #2 | 5/64 | 1.98 | +0.14 0 | 3/16 | 4.76 | 5/64 | 1.98 | 4.4 | 2.54 | 12.6 |
| | 033102 | I9MT12T2A3-NC2033 | | | #3 | 7/64 | 2.78 | | 1/4 | 6.35 | 7/64 | 2.78 | 5.9 | | 13.8 |
| | 033103 | I9MT12T2A4-NC2033 | | | #4 | 1/8 | 3.18 | +0.18 0 | 5/16 | 7.94 | 1/8 | 3.18 | 7.3 | | 14.25 |
| 16 | 034101 | I9MT1603A5-NC2033 | | | #5 | 3/16 | 4.76 | | 7/16 | 11.11 | 3/16 | 4.76 | 10.3 | 3.18 | 20.0 |
| | 035101 | I9MT2004A6-NC2033 | | | #6 | 7/32 | 5.56 | | 1/2 | 12.7 | 7/32 | 5.56 | 11.8 | 4.76 | 27.75 |
| 20 | 035102 | I9MT2004A7-NC2033 | | | #7 | 1/4 | 6.35 | +0.22 0 | 5/8 | 15.88 | 1/4 | 6.35 | 14.6 | | 28.5 |
| | 035103 | I9MT2004A8-NC2033 | | | #8 | 5/16 | 7.94 | | 3/4 | 19.05 | 5/16 | 7.94 | 17.6 | | 29.0 |
| 25 | 036101 | I9MT2506A10-NC2033 | | | #10 | 3/8 | 9.53 | | 0.98" | 25.0 | 3/8 | 9.53 | 22.9 | 6.35 | 34.9 |

## Measuring Master >>

- Apply on lathe to align the center of work spindle and tool.
- Each insert has just one measuring tip.
- Concentricity: ±0.01mm




| IC | Parts No. | d1 | L | L1 |
|---|---|---|---|---|
| 08 | I9MT08T1-MM | Ø1.9 | 9.0 | 3.0 |
| 10 | I9MT1003-MM | Ø2.9 | 11.5 | 5.0 |
| 12 | I9MT12T2-MM | Ø2.3 | 13.5 | 3.5 |
| 16 | I9MT1603-MM | Ø3.0 | 19.0 | 5.0 |
| 20 | I9MT2004-MM | Ø4.5 | 26.0 | 8.0 |

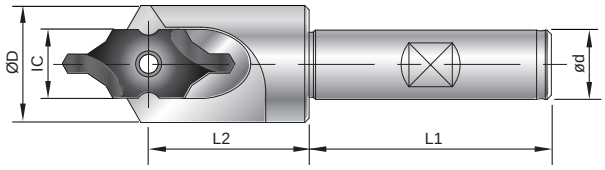


# Holders of Indexable Center Drill

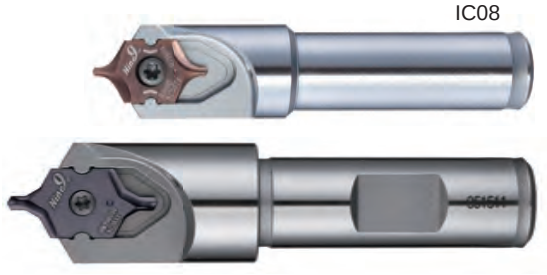



## Holder >>

- Made of hardened high alloy steel, HRC 53.
- IC08 shank is cylindrical shank.
  Other shanks are weldon shank.




IC08

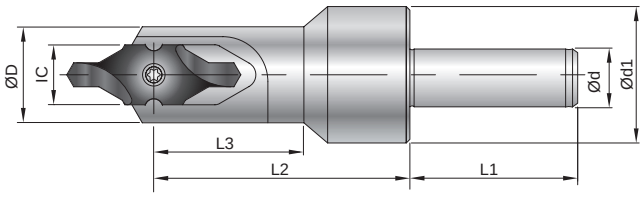


| IC | Code | Parts No. | Type | ød | L1 | L2 | ØD | Screw | Key |
|---|---|---|---|---|---|---|---|---|---|
| 08 | 802002 | 00-99616-IC08-10F | BC10-IC08F | 10 | 30 | 18.5 | 12 | *NS-25060 0.9 Nm | NK-T7 |
| | 812002 | 00-99616-IC08-3/8F | BC3/8"-IC08F | 3/8" | | | | | |
| 10 | 801002 | 00-99616-IC10-12F | SB12-IC10F | 12 | 45 | 24.5 | 16 | *NS-25060 0.9 Nm | NK-T7 |
| 12 | 803002 | 00-99616-IC12-16F | SB16-IC12F | 16 | 48 | 30.5 | 21 | NS-30072 2.0 Nm | NK-T9 |
| | 813002 | 00-99616-IC12-5/8F | SB5/8"-IC12F | 5/8" | | | | | |
| 16 | 804002 | 00-99616-IC16-16F | SB16-IC16F | 16 | 48 | 37 | 27 | NS-35080 2.5 Nm | NK-T15 |
| | 814002 | 00-99616-IC16-5/8F | SB5/8"-IC16F | 5/8" | | | | | |
| 20 | 805002 | 00-99616-IC20-20F | SB20-IC20F | 20 | 50 | 51 | 32 | NS-50125 5.5 Nm | NK-T20 |
| | 815002 | 00-99616-IC20-3/4F | SB3/4"-IC20F | 3/4" | | | | | |
| 25 | 806002 | 00-99616-IC25-25F | SB25-IC25F | 25 | 56 | 56 | 43 | NS-50125 5.5 Nm | NK-T20 |
| | 816002 | 00-99616-IC25-1F | SB 1"-IC25F | 1" | | | | | |

*Torque screwdriver is recommended.

## Cylindrical Shank with Pre-balanced >>

- Pre-balanced holder enhance the stability of centering to get high accurate profile.
- G6.3 / 10,000 r.p.m.

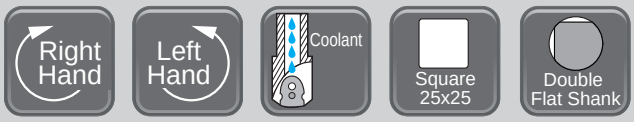





| IC | Code | Parts No. | Type | Ød | Ød1 | L1 | L2 | L3 | ØD | Screw | Key |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 08 | 802003 | 00-99616-IC08-10B | BC10-IC08B | 10 | 22 | 30 | 33.5 | 19 | 12 | *NS-25060 0.9 Nm | NK-T7 |
| 12 | 803003 | 00-99616-IC12-12B | BC12-IC12B | 12 | 34 | 48 | 51 | 30 | 21 | NS-30072 2.0 Nm | NK-T9 |
| 16 | 804003 | 00-99616-IC16-16B | BC16-IC16B | 16 | 39 | 48 | 67 | 37 | 27 | NS-35080 2.5 Nm | NK-T15 |
| 20 | 805003 | 00-99616-IC20-20B | BC20-IC20B | 20 | 49 | 50 | 86 | 51 | 32 | NS-50125 5.5 Nm | NK-T20 |
| 25 | 806003 | 00-99616-IC25-25B | BC25-IC25B | 25 | 59 | 56 | 99 | 56 | 43 | NS-50125 5.5 Nm | NK-T20 |

*Torque screwdriver is recommended.

# Holders of Indexable Center Drill



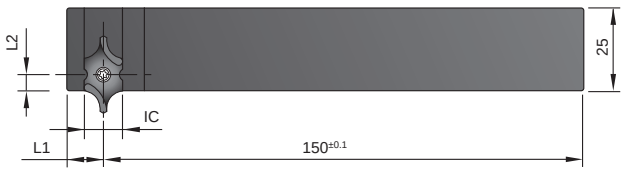






## Square Shank 25 x 25 Right / Left Hand >>

- For used on lathe, clamp by VDI and BMT holders.
- Made of hardened alloy steel, HRC 40.
- Other sizes are available on request.



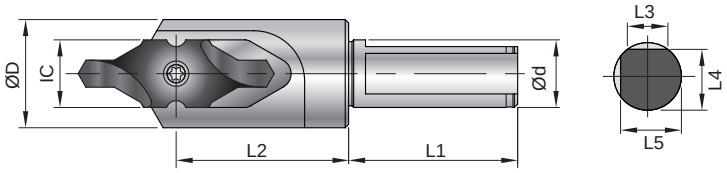



| IC | Code | Parts No. | L1 | L2 | Screw | Key |
|---|---|---|---|---|---|---|
| 08 | 822022 | 00-99616-IC08-R2525MF | 8 | 3.25 | *NS-25060 0.9 Nm | NK-T7 |
| | 822012 | 00-99616-IC08-L2525MF | | | | |
| 12 | 823022 | 00-99616-IC12-R2525MF | 11 | 4.9 | NS-30072 2.0 Nm | NK-T9 |
| | 823012 | 00-99616-IC12-L2525MF | | | | |
| 16 | 824022 | 00-99616-IC16-R2525MF | 13 | 4.9 | NS-35080 2.5 Nm | NK-T15 |
| | 824012 | 00-99616-IC16-L2525MF | | | | |

*Torque screwdriver is recommended.

## Double Flat Shank >> Non-Stock Item

- Used on lathe.
- Double flat shank design for tool holder with side lock flat.
- 180° for insert at top, 90° for insert at front.




| IC | Code | Parts No. | Type | Ød | L1 | L2 | L3 | L4 | L5 | ØD | Screw | Key |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 08 | 802004 | 00-99616-IC08-10S | SL10-IC08S | 10 | 30 | 18.5 | 6 | 9 | 9 | 12 | *NS-25060 0.9 Nm | NK-T7 |
| 12 | 803004 | 00-99616-IC12-16S | SL16-IC12S | 16 | 48 | 30.5 | 9.33 | 14.5 | 14.5 | 21 | NS-30072 2.0 Nm | NK-T9 |
| 16 | 804004 | 00-99616-IC16-16S | SL16-IC16S | 16 | 48 | 37 | 9.33 | 14.5 | 14.5 | 27 | NS-35080 2.5 Nm | NK-T15 |
| 20 | 805004 | 00-99616-IC20-20S | SL20-IC20S | 20 | 50 | 51 | 12 | 18 | 18 | 32 | NS-50125 5.5 Nm | NK-T20 |
| 25 | 806004 | 00-99616-IC25-25S | SL25-IC25S | 25 | 56 | 56 | 13.57 | 23 | 23 | 43 | NS-50125 5.5 Nm | NK-T20 |

*Torque screwdriver is recommended.

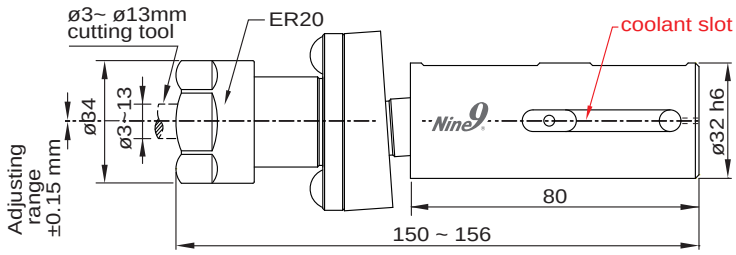
# Center Height Adjusting Sleeve

## Principle >>

- Designed for adjusting Center Height of center drills, NC spot drills, reamers and taps on the CNC lathes.
- The main body is made from two sleeves. The inner sleeve is to hold and lock the cutting tool.
- Its center is inclined to the outer sleeve. When the inner sleeve is pushed or pulled, the cutting tool's center height is adjusted to lower or higher position.



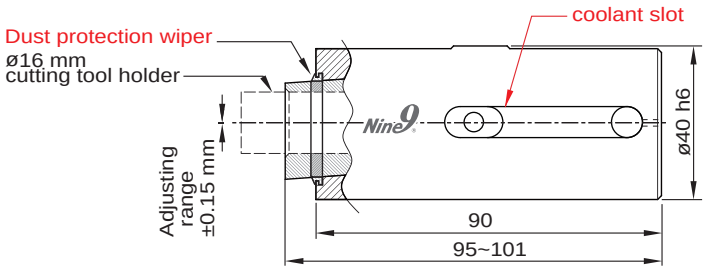
## Parts No.:00-99600-320H >>






Type : SB32-IDER20

## Parts No.:00-99600-400H >>

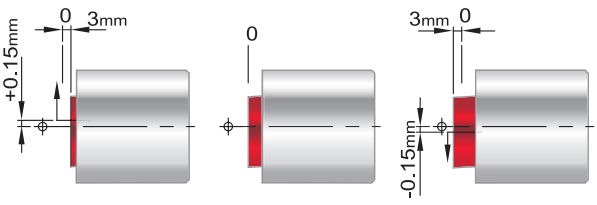


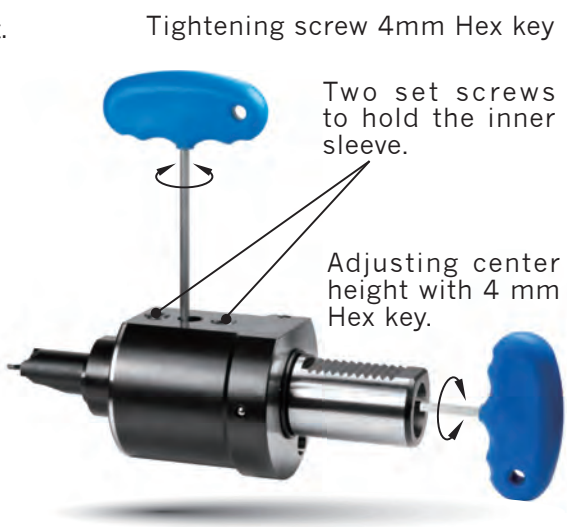



Type : SB40-ID16

## Applications >>

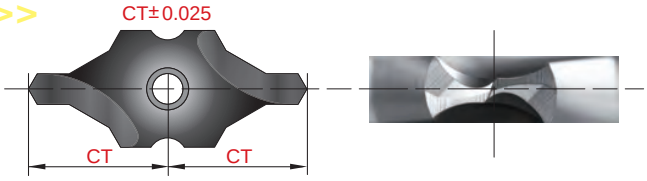
- Used when the CNC lathes need to adjust the center height.
- This sleeve can be clamped by VDI 40, VDI 50 E2 tool holders,and other types internal turning tool holders.
- Center height adjusting range: ±0.15 mm(.006").
- Total axial movement is 6mm(.236").



## Profit by making the right choice >>

- High speed and feed rate reduce cutting time.
- The unique design increases tool life and reduces change over time.

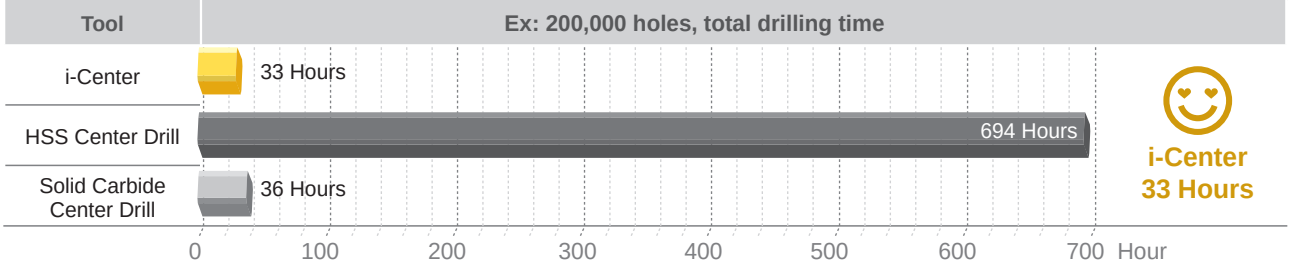


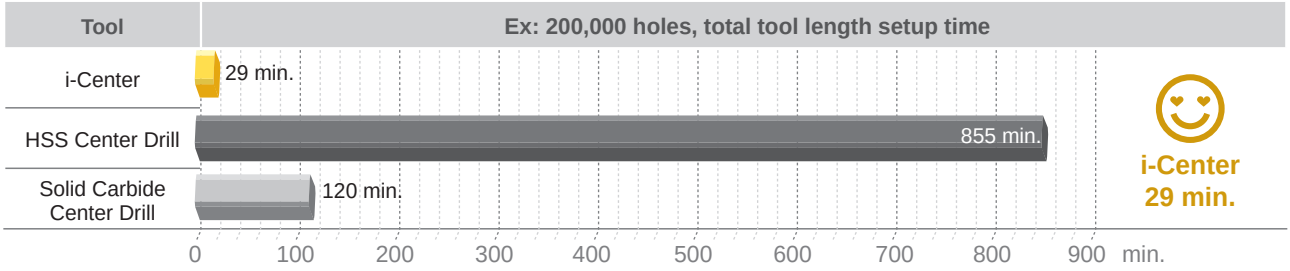

▲ 2 cutting edges ▲ 2 cutting flutes design

## Comparison >>

- Workpiece : Low carbon alloy steel, 850 N/mm$^2$
- Machine: VMC BT40 with internal coolant

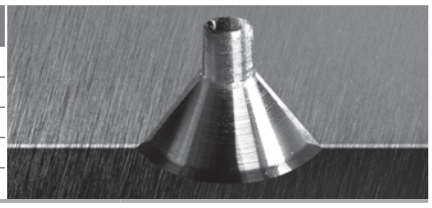
| Diameter of tool : Ø3.15 mm<br>Depth of drilling : 7.2 mm | |  |  |  |
|---|---|---|---|---|
| **Comparison** | | **i-Center** | **HSS Center Drill ( TiN Coating )** | **Solid Carbide Center Drill** |
| **Cutting speed** | **m/min.** | 65 | 17 | 65 |
| **Spindle speed** | **r.p.m.** | 6570 | 1718 | 6570 |
| **Feed rate f =** | **mm/rev.** | 0.12 | 0.02 | 0.1 |
| **Feed rate F=** | **mm/min.** | 788.4 | 34.4 | 657 |
| **Coolant** | **Emulsion** | External / Internal | External | External |
| **Drilling time** | **sec.** | 0.55 | 12.5 | 0.65 |
| **Holes of drilling per edge** | | 7000 | 700 | 5000 |

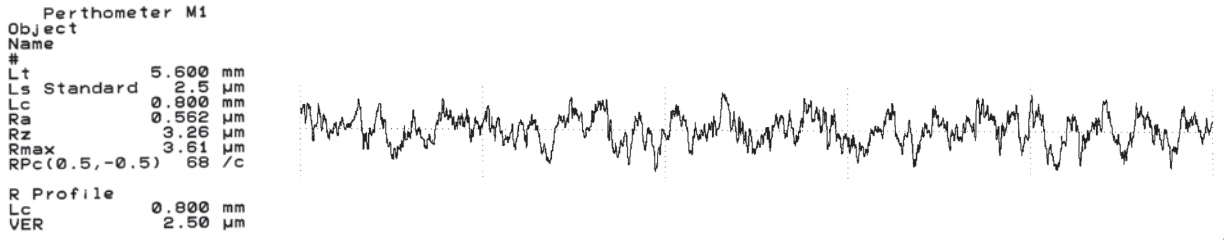



## Surface finish >>

| i-Center Insert | Material SCM440 | | |
|---|---|---|---|
| **I9MT1603B0500**<br>**NC2033** | **Vc** | 60 | m/min. |
| | **S** | 3800 | r.p.m. |
| | **f** | 0.1 | mm/rev. |
| | **F** | 380 | mm/min. |
| | **Ap** | 13.5 | mm |



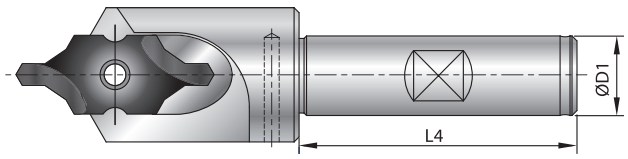


# i-Center Enquiry Form

The following information should be checked while discussing with customer.

| | | |
|---|---|---|
| **Machine** | | |
| Machine Type | | |
| Spindle Speed | Max. | r.p.m. |
| Power of Spindle motor | ☐ KW | ☐ HP |
| Coolant supply | ☐ NO | |
| | ☐ If yes, ☐ External | |
| | ☐ Internal | bar(psi) |
| **Current tool** | | |
| Cutting Speed | ☐ HSS | ☐ Solid Carbide |
| | m/min. | SFM |
| Others | | |
| Feed Rate | mm/rev. | inch/rev. |
| **Work Piece** | | |
| Material code | | |
| Center hole type | ☐ R ☐ A ☐ B ☐ C | |
| | ☐ Other as attached drawing | |
| Other request | ☐ Surface roughness | |
| | ☐ Tolerance(see below) | |

## Special Tool holder shank dimensions >>



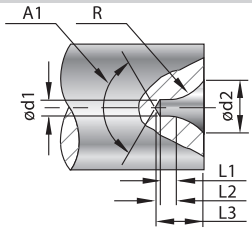

| Dimension Table | A1 | A2 | A3 | ød1 | ød2 | ød3 |
|---|---|---|---|---|---|---|
| Dimension | | | | | | |
| Tolerance | --- | +0° −1° | ±1° | ±0.05 | ±0.05 | --- |

| Dimension Table | L1 | L2 | L3 | R | øD1 | L4 |
|---|---|---|---|---|---|---|
| Dimension | | | | | | |
| Tolerance | ±0.05 | ±0.05 | ±0.05 | ±0.5 | h6 | --- |

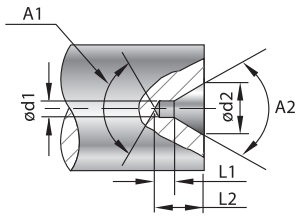
## Center hole dimension >>

- Please provide workpiece drawing
- One of following type should be chosen.

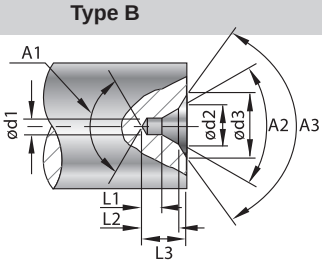
**Type R**



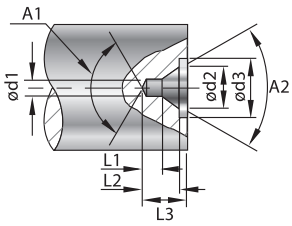

**Type A**



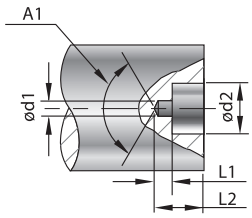

**Type B**




**Type C**



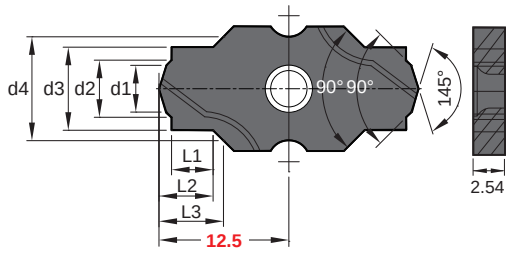

**Other**




## DIN332 Form D >> Non-Stock Item



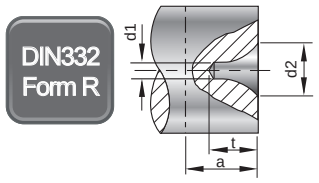
- M3, M4 Counterbore Insert, K20F grade, TiAlN coated.

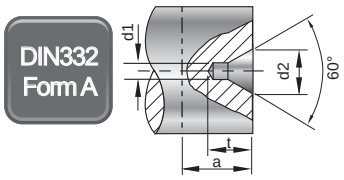


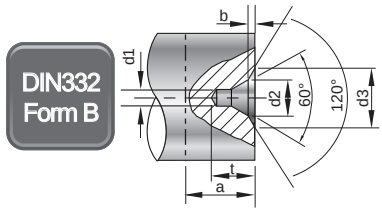

| Part No. | d1 | d2 | d3 | d4 | L1 | L2 | L3 |
|---|---|---|---|---|---|---|---|
| I9MT12T2M03-NC2033 | 3.40 | 4.40 | 6.00 | 8.00 | 3.00 | 4.04 | 5.04 |
| I9MT12T2M04-NC2033 | 4.50 | 5.50 | 8.00 | 10.0 | 4.00 | 5.21 | 6.21 |

# Technical Standard
# ISO 2541-1972 / DIN332

## 60° Center holes >>



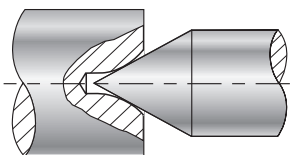
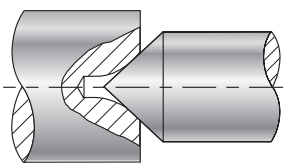
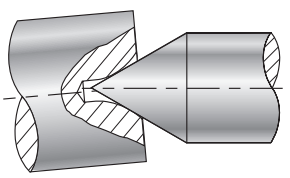



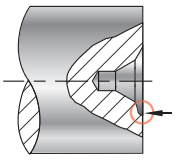
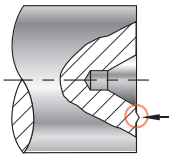
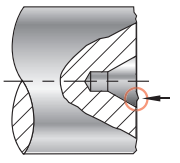

| STD | DIN332 Form R ISO 2541-1972 | | | DIN332 Form A ISO 866-1975 | | | DIN332 Form B ISO 2540 1973 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| d1 | d2 | t | a | d2 | t | a | d2 | b | d3 | t | a |
| 1 | 2.12 | 1.9 | 3 | 2.12 | 1.9 | 3 | 2.12 | 0.3 | 3.15 | 2.2 | 3.5 |
| 1.25 | 2.65 | 2.3 | 4 | 2.65 | 2.3 | 4 | 2.65 | 0.4 | 4 | 2.7 | 4.5 |
| 1.6 | 3.35 | 2.9 | 5 | 3.35 | 2.9 | 5 | 3.35 | 0.5 | 5 | 3.4 | 5.5 |
| 2 | 4.25 | 3.7 | 6 | 4.25 | 3.7 | 6 | 4.25 | 0.6 | 6.3 | 4.3 | 6.6 |
| 2.5 | 5.3 | 4.6 | 7 | 5.3 | 4.6 | 7 | 5.3 | 0.8 | 8 | 5.4 | 8.3 |
| 3.15 | 6.7 | 5.8 | 9 | 6.7 | 5.9 | 9 | 6.7 | 0.9 | 10 | 6.8 | 10 |
| 4 | 8.5 | 7.4 | 11 | 8.5 | 7.4 | 11 | 8.5 | 1.2 | 12.5 | 8.6 | 12.7 |
| 5 | 10.6 | 9.2 | 14 | 10.6 | 9.2 | 14 | 10.6 | 1.6 | 16 | 10.8 | 15.6 |
| 6.3 | 13.2 | 11.4 | 18 | 13.2 | 11.5 | 18 | 13.2 | 1.4 | 18 | 12.9 | 20 |
| 8 | 17 | 14.7 | 22 | 17 | 14.8 | 22 | 17 | 1.6 | 22.4 | 16.4 | 25 |
| 10 | 21.2 | 18.3 | 28 | 21.2 | 18.4 | 28 | 21.2 | 2 | 28 | 20.4 | 31 |

* a: Minimum material will be cut. If the center hole will be removed after turning or grinding. (mm/inch)

## Form R center hole apply for >>

| 60° Center of tail stock | 90° Center of tail stock | Center hole and center are misaligned |
|---|---|---|
|  |  |  |

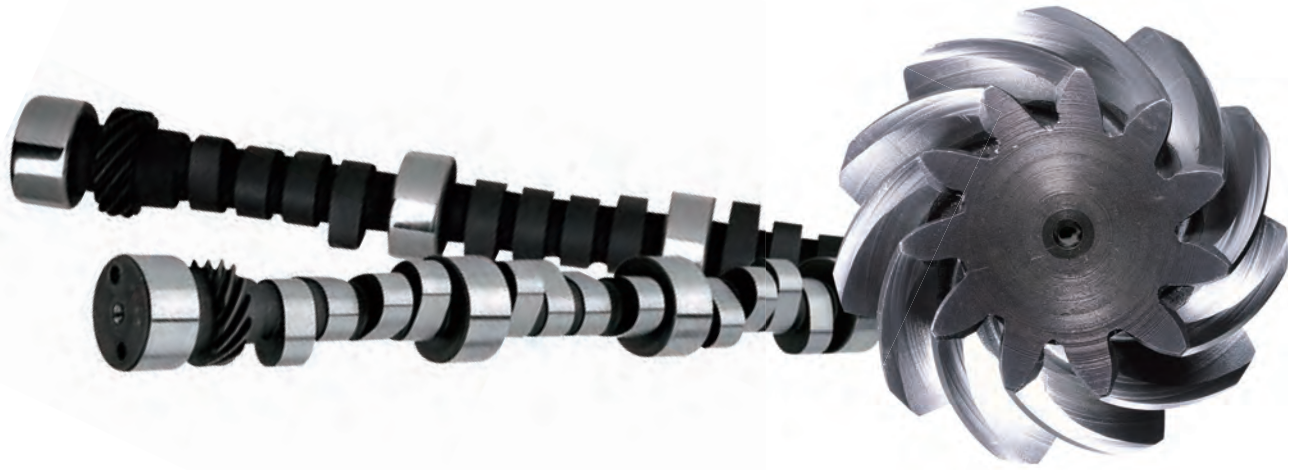
## Form B center hole apply for >>

| Avoid scar or distortion while transportation | Burr | Rough surface of workpiece |
|---|---|---|
|  |  |  |

# i-Center Applications

## Tip >>

- Various centering applications and products - shafts of engine, transmission gear, bearings, motors, grinding parts, spindles, gear reducers, cooling fan, universal joints...
- Special forms for other applications also available on request.

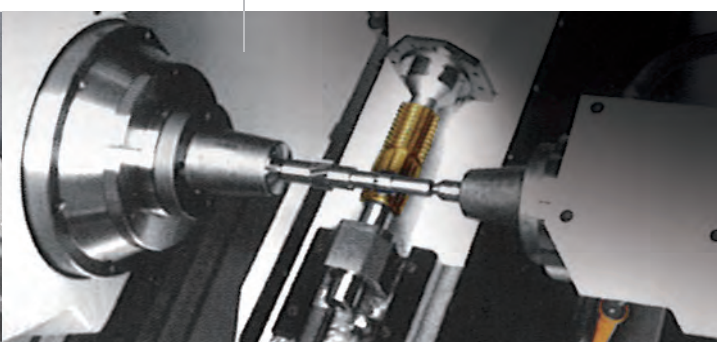










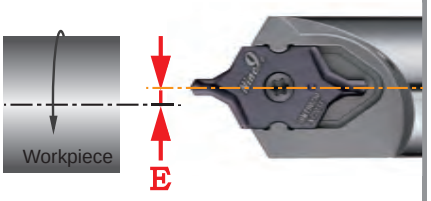


# Technical Guide

## Before you start, please pay attention the following conditions >>

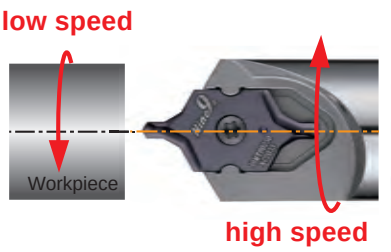
### Center misalignment

**E** must be < 0.02mm.



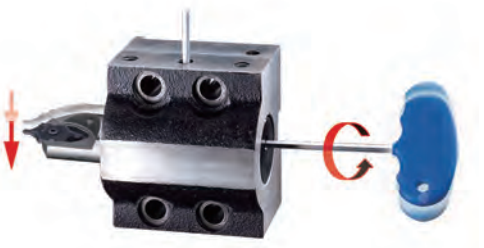

### Driving Tool

low speed : high speed
**1 : 3**




### Center height adjusting sleeve

When CNC lathe turret center is misaligned ≥ 0.15mm, please use center height adjusting sleeve. (See page 61)



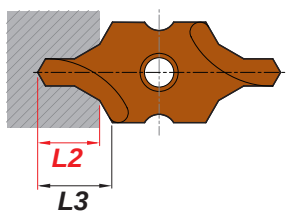
### Internal coolant

Internal coolant is recommended.

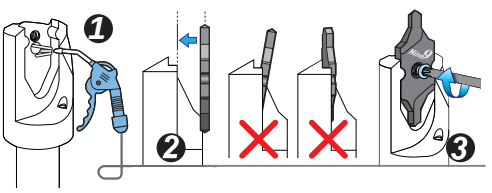


### DIN 332 Form A+B

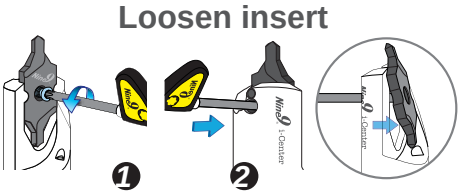
Reduce 30% of Spindle speed and keep same feed rate (mm/rev.) while depth L2 is reached.




### Clamping insert




**Loosen insert**




| Metric | | Inch | |
|---|---|---|---|
| $S = \frac{Vc \times 1000}{\pi \times d1}$ | **d1 =** diameter -mm | $S = \frac{(3.82 \times SFM)}{d1}$ | **d1 =** diameter-inch |
| | **S =** Spindle Speed -r.p.m. | | **S =** Spindle Speed-r.p.m. |
| | **Vc =** Cutting Speed -m/min. | | **SFM =** Surface Speed-ft./min. Vc (m/min.) x 3.28 |
| **F = S x f** | **f =** mm/rev. | **F = IPRxr.p.m** | **f = IPR =** inch/rev. |
| | **F =** mm/min. | | **F =** inch/min. |

# Cutting Data

## Ø1~Ø3.15 (A2~A4)

| Workpiece material | | Vc (m/min.) | d1 (pilot dia.) | IC08 / IC10 | | IC12 | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Ø1~1.25 | Ø1.6~3.15 | Ø2 (A2) | Ø2.5 (A3) | Ø3.15 (A4) |  |  |
| P | Carbon steel C<0.3% | < 80 | S r.p.m. | 2000 ~ 10000 | 1600 ~ 8000 | 1600 ~ 8000 | 1400 ~ 7000 | 1200 ~ 6000 | ● | ○ |
| | | | f mm/rev. | 0.02~0.03~0.05 | 0.03~0.05~0.06 | 0.04~0.06~0.08 | 0.06~0.08~0.10 | 0.08~0.10~0.12 | | |
| P | Carbon steel C>0.3% | < 70 | S r.p.m. | 2000 ~ 9000 | 1600~ 7200 | 1600 ~ 7200 | 1400 ~ 6300 | 1200 ~ 5400 | ● | ○ |
| | | | f mm/rev. | 0.02~0.03~0.05 | 0.03~0.04~0.05 | 0.03~0.04~0.05 | 0.06~0.08~0.10 | 0.08~0.10~0.12 | | |
| P | Low alloy steel C<0.3% | < 65 | S r.p.m. | 2000 ~ 8000 | 1600 ~ 6400 | 1600 ~ 6400 | 1400 ~ 5600 | 1200 ~ 4800 | ● | ○ |
| | | | f mm/rev. | 0.01~0.02~0.04 | 0.02~0.03~0.05 | 0.02~0.03~0.05 | 0.04~0.06~0.08 | 0.06~0.08~0.10 | | |
| P | High alloy steel | < 60 | S r.p.m. | 1000 ~ 6000 | 800 ~ 4800 | 800 ~ 4800 | 700 ~ 4200 | 600 ~ 3600 | ● | ○ |
| | | | f mm/rev. | 0.01 ~ 0.02 | 0.01~0.02~0.04 | 0.01~0.02~0.04 | 0.02~0.04~0.06 | 0.04~0.06~0.08 | | |
| M | Stainless steel | < 20 | S r.p.m. | 1000 ~ 3000 | 800 ~ 2400 | 800 ~ 2400 | 700 ~ 2100 | 600 ~ 1800 | ● | ○ |
| | | | f mm/rev. | 0.003 ~ 0.01 | 0.005 ~ 0.02 | 0.01 ~ 0.02 | 0.01~0.02~0.03 | 0.02~0.03~0.05 | ≥ 5 bar | |
| K | Cast Iron | < 70 | S r.p.m. | 2000 ~ 9000 | 1600 ~ 7200 | 1600 ~ 7200 | 1400 ~ 6300 | 1200 ~ 5400 | Air | |
| | | | f mm/rev. | 0.01~0.02~0.04 | 0.02~0.04~0.06 | 0.02~0.04~0.06 | 0.04~0.06~0.08 | 0.06~0.08~0.10 | | |
| N | Non-ferrous metal | < 200 | S r.p.m. | 6000 ~ 20000 | 4800 ~ 16000 | 4800 ~ 16000 | 4200 ~ 14000 | 3600 ~ 12000 | ● | ○ |
| | | | f mm/rev. | 0.01~0.02~0.03 | 0.01~0.02~0.04 | 0.01~0.02~0.04 | 0.02~0.03~0.05 | 0.02~0.04~0.06 | | |

● Best ○ Possible

## Ø4~Ø10 (A5~A10)

| Workpiece material | | Vc m/min. | d1 (pilot dia.) | IC16 | | IC20 | | IC25 |  |  |
|---|---|---|---|---|---|---|---|---|---|---|
| | | | | Ø4 (A5) | Ø5 (A6) | Ø6.3 (A7) | Ø8 (A8) | Ø10 (A10) | | |
| P | Carbon steel C<0.3% | < 80 | S r.p.m. | 1000 ~ 5000 | 900 ~ 4500 | 800 ~ 4000 | 700 ~ 3500 | 600 ~ 3000 | ● | ○ |
| | | | f mm/rev. | 0.08~0.12~0.14 | 0.10~0.12~0.16 | 0.10~0.14~0.16 | 0.12~0.15~0.18 | 0.14~0.18~0.20 | | |
| P | Carbon steel C>0.3% | < 70 | S r.p.m. | 1000 ~ 4500 | 900 ~ 4050 | 800 ~ 3600 | 700 ~ 3150 | 600 ~ 2700 | ● | ○ |
| | | | f mm/rev. | 0.08~0.12~0.14 | 0.10~0.12~0.16 | 0.10~0.14~0.16 | 0.12~0.15~0.18 | 0.14~0.18~0.20 | | |
| P | Low alloy steel C<0.3% | < 65 | S r.p.m. | 1000 ~ 4000 | 900 ~ 3600 | 800 ~ 3200 | 700 ~ 2800 | 600 ~ 2400 | ● | ○ |
| | | | f mm/rev. | 0.06~0.08~0.10 | 0.08~0.10~0.12 | 0.08~0.12~0.14 | 0.10~0.14~0.16 | 0.12~0.16~0.20 | | |
| P | High alloy steel | < 60 | S r.p.m. | 500 ~ 3000 | 450 ~ 2700 | 400 ~ 2400 | 350 ~ 2100 | 300 ~ 1800 | ● | ○ |
| | | | f mm/rev. | 0.04~0.06~0.08 | 0.06~0.08~0.10 | 0.08~0.10~0.12 | 0.10~0.14~0.16 | 0.10~0.14~0.16 | | |
| M | Stainless steel | < 25 | S r.p.m. | 500 ~ 1500 | 450 ~ 1350 | 400 ~ 1200 | 350 ~ 1050 | 300 ~ 900 | ● | ○ |
| | | | f mm/rev. | 0.02~0.04~0.06 | 0.02~0.04~0.06 | 0.04~0.06~0.08 | 0.04~0.06~0.08 | 0.05~0.07~0.10 | ≥ 5 bar | |
| K | Cast Iron | < 70 | S r.p.m. | 1000 ~ 4500 | 900 ~ 4050 | 800 ~ 3600 | 700 ~ 3150 | 600 ~ 2700 | Air | |
| | | | f mm/rev. | 0.06~0.08~0.10 | 0.08~0.10~0.12 | 0.08~0.12~0.14 | 0.10~0.14~0.16 | 0.12~0.16~0.18 | | |
| N | Non-ferrous metal | < 200 | S r.p.m. | 3000 ~ 10000 | 2700 ~ 9000 | 2400 ~ 8000 | 2100 ~ 7000 | 1800 ~ 6000 | ● | ○ |
| | | | f mm/rev. | 0.02~0.04~0.06 | 0.04~0.06~0.08 | 0.04~0.06~0.08 | 0.06~0.08~0.10 | 0.06~0.08~0.10 | | |

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